



Midterm Examinations

1st Semester, S.Y. 2024 – 2025
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HAU School of Computing
First Year | Bachelor of Science in
Computer Science

Syllabus:

- 1. Introduction to Statistics
 - 1.1. Areas of Statistics
 - 1.2. Sampling Methods
 - 1.3. Level or Scales of Measurements
 - 1.4. Collection of Data
 - 1.4.1. Methods of Collecting Data
 - 1.5. Methods of Presenting Data
 - 1.5.1. Using Graphs and Tables
- 2. Inferential Statistics
 - 2.1. Hypothesis Testing
- 3. Problem Solving
 - 3.1. Polya’s Four-Step Process for Problem Solving
 - 3.2. Approaches to Problem Solving
- 4. Mathematical Investigation
 - 4.1. Processes Involved in an Investigation
 - 4.2. Conjectures

**Should there be any mistakes, inconsistencies, or inaccuracies in this material, please feel free to reach out to the transcriber. Good luck and study well for your exams!*

Last updated: September 8, 2024 at 3:13 AM

Introduction to Statistics

Statistics

- Is a branch of mathematics that deals with the scientific collection, organization, presentation, analysis and interpretation of data to obtain useful and meaningful information.

Data

- Any piece of information useful to the researcher; the measurements obtained in a research study.

Individuals

- Are the people, places, events, or objects included in the study.

Variable

- Is a characteristic or property of an individual to be measured or observed.

a. Quantitative Variable

- Has a value or numerical measurement for which operations such as addition or averaging make sense.
- It is a variable that **quantifies** an element of a population.
- Examples: distance travelled to school, number of children in a family, monthly electric bill, etc.
 - **Discrete Variable** – is a variable that can be obtained by **counting**.
 - **Continuous Variable** – is a variable that can be obtained by **measuring**.

b. Qualitative Variable

- A variable that categorizes or describes an element of a population.
- It describes an individual by placing the individual into a category or group such as male or female. Sometimes, they are referred to as **categorical** variables.

- Examples: gender, hair color, ethnicity, civil status, etc.

Population

- Refers to the **totality of all the individuals** which one has an interest in at a particular time.
- It refers to the entire group of individuals or objects known to have similar characteristics.
- **Parameter** – a value that describes an aspect of a population.

Sample

- It is a **part of a population** determined by sampling procedures.
- It is a **subset** of the entire population. It refers to one or more elements taken from the population for a specific purpose.
- **Statistic** – a value that describes an aspect of a sample.

Areas of Statistics

A. Descriptive Statistics

- Methods concerned with collecting, describing (organizing, presenting, summarizing, and analyzing a set of data **without drawing conclusions or inferences** about a large group.
- Examples:
 - A teacher computes the average grade of his students and then determines the top 10 students.
 - A janitor counts the number of various furniture inside the school

B. Inferential Statistics

- Methods concerned with drawing conclusions, making predictions, or forecasts, about the entire set of data.
- Examples:
 - A manager of a business firm predicts future sales of the company based on the present sales.
 - A psychologist investigates if there is a significant relationship between the mental age and chronological age.
 - A researcher studies the effectiveness of a new fertilizer in increasing food production.

Sampling Methods

A. Probability Sampling (Random)

- Each member of the population has an equal chance of being selected.

1. Simple Random Sampling

- Also known as “**lottery method**” or the “**fishbowl technique**”, the most commonly used sampling technique.
- Every element of the population has the same probability of being selected for the inclusion in the sample.
- Example: A company executive selected 15 participants for the international business Congress by writing the names of all employees and then choosing a name one at a time until the delegation is completed.

2. Systematic Sampling

- Adopts a **skipping pattern** in the selection of sample units.



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- Every **n**th element of the population is selected until the desired number of elements in the sample is obtained.
- Example: From a list containing the names of 500 members of the alumni association, a sample of size 50 is obtained by including **every 10th person** in the list.

3. Stratified Sampling

- The population is partitioned into subgroups called **strata**, based on some characteristics like year, religion, gender, age, ethnicity, etc.
- Then, samples are randomly selected separately in each stratum.

4. Cluster Sampling

- The population is randomly divided into groups (or **clusters**).
- It is used when “natural” but relatively **heterogeneous** groupings are evident in a statistical population.
- Example: Suppose a researcher wants to study the effect of blended learning in a particular city/municipality. Since there are many schools in a city/municipality, it will be very inefficient and impractical to consider all the schools in the study. Instead, the researcher will randomly choose few schools where students will be surveyed.

B. Non-probability Sampling (Nonrandom)

1. Convenience Sampling

- It is also known as **grab** or **opportunity sampling**, or **accidental** or **haphazard** sampling.
- It involves the sample being drawing that part of the population which is close to hand. That is, readily available and convenient.
- Example: The researcher conducts an online poll in social media in determining the respondent's choice of TV program.

2. Purposive Sampling

- The samples are selected with a purpose in mind.
- It is also called **judgmental** or **selective sampling**, as they focus on samples which are taken based on the judgment of the researcher.
- Example: The teacher announced that he needs three (3) representatives to represent the school in a regional quiz show. The teacher decided to choose the top three students as representatives.

3. Snowball Sampling

- It is sometimes called **chain-referral** sampling.
- The researcher chooses a possible respondent for the study at hand. Then, each respondent is asked to give recommendations or referrals to other possible respondents.
- It is used when the research is focused on participants with **very specific characteristics** such as being members of a gang.
- Example: The researcher asks respondents to suggest other possible respondents who are members of their organization.

4. Quota Sampling

- The researcher starts by identifying quotas, which are predefined control categories such as age, gender, education, or religion.

Level or Scales of Measurements

1. Nominal Level of Measurement

- It applies to data that consists of **names, labels, or categories**.
- There are no implied criteria by which the data can be ordered from smallest to highest.
- Examples: gender, sports, blood type, civil status, brand of vaccine injected, etc.

2. Ordinal Level of Measurement

- It applies to data that can be arranged in **order** or **rank**.
- The differences between the data values either cannot be determined or are meaningless.
- Examples: ranking of students, military ranks, sizes of shirt, etc.

3. Interval Level of Measurement

- It applies to data that can be arranged in order.
- Differences between data values are meaningful.
- Data at this level have no true, or meaningful zero.
- Examples: Temperature in degree Celsius, test results, grades, IQ, etc.

4. Ratio Level of Measurement

- It applies to data that can be arranged in order.
- Both differences between data values, and ratio of data values are meaningful.
- Data at the ratio level have a true or meaningful zero.
- Example: area of a circle, distance travelled, volume of water, etc.

Collection of Data

- This is the first step in conducting a study or research after the formulation of the research problem.
- Compilation of primary and secondary sources of information
- The integration of different sources will consolidate the writeup of the report.

Classifications of Data

- a. **Primary Data** – are first-hand information that is gathered directly from the source, such through personal interviews.
- b. **Secondary Data** – are data that have been collected for another purpose. These are data derived from books, encyclopedias, journals, magazines, and other research or studies conducted by other individuals.

Methods of Collecting Data

1. Interview Method

- Referred to as **direct method**
- It involves an interviewer and an interviewee.
- Advantage/s: The researcher can get more accurate answers or responses since clarifications can be made during the interview.
- Disadvantage/s: It can be time-consuming and costly.

2. Questionnaire Method

- Referred to as **indirect method**
- The questionnaire (also called **survey**) is a set of questions given to a sample of people.



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- Advantage/s: It can save time and money and it can cover a lot of respondents.
- Disadvantage/s: Questions can't be clarified; return rate of questionnaire can be low

3. Observation Method

- Making use of the different human senses in gathering information.
- Advantage/s: Data collected by observation, are, thus, more objective and generally more accurate.
- Disadvantage/s: Limiting factor to observe; time-consuming and costly

4. Registration or Census

- Governed by enacted laws
- Covers a large scope of population or the entire population
- Some government agencies are Land Transportation Office (LTO), Commission on Elections (COMELEC) and Philippine Statistics Authority (PSA)

5. Experimental Method

- Involves performing an experiment and then collecting the data to be further analyzed.
- There are two types of experiments – laboratory and field.
- It is used when establishing **cause-and-effect** relationships.

Methods of Presenting Data

- Visual methods can make the point much stronger than simply describing the data.
- Appropriate use of tables and graphs can enhance the message you are delivering, but they have the potential to confuse or convey wrong messages.
- Always be sure to explain your table or graph in the text of your report. For every table or graph, there should be a reference to it in the text.

Summary Statistics

- It is a part of descriptive statistics that summarizes and provides the gist of information about the sample data.

a. Measures of Central Tendency

1. **Mean** – arithmetic average; it is the most common and most useful average

Mean of a Population

$$\mu_x = \frac{\sum X}{N}$$

Mean of a Sample

$$\bar{X} = \frac{\sum X}{n}$$

Where:

$\bar{X}$ = sample mean

n = sample size

μ = population mean

N = population size

2. **Median** – middlemost value

3. **Mode** – most frequently occurring value

b. Measures of Variability

1. **Range** – distance from the lowest to the highest value (using two data points)
2. **Variance** – uses all data points; it is the average distance that each score is from the mean.
3. **Standard deviation**

Presentation of Data Using Graphs and Tables

1. Pie Chart

- A pie chart is a circular chart divided into sectors, illustrating relative magnitudes usually in percentages. In a pie chart, the area is proportional to the quantity it represents.

2. Line Graph

- A line graph (also known as a line plot or line chart) is a graph which uses lines to connect individual data points that display quantitative values over a specified time interval.
- Line graphs use data point markers that are connected by straight lines to aid in visualization.

3. Bar Graph

- A **bar chart** is a graph with rectangular bars. The graph usually compares different categories.
- Although the graphs can be plotted vertically or horizontally, the most usual type of bar graph is **vertical**.
- a. **Grouped Bar Graph** – it is a way to show information about sub-groups of the main categories.
- b. **Stacked Bar Chart** – it also shows sub-groups, but the sub-groups are stacked on the same bar.
- c. **Segmented Bar Graph** – each bar shows 100% of the discrete value.

4. Histogram

- A histogram is a display of statistical information that uses rectangles to show the frequency of data items in successive numerical intervals of equal size.
- In the most common form of histogram, the **independent variable** is plotted along the **horizontal axis** and the **dependent variable** is along the **vertical axis**.

5. Scatterplots

- A scatterplot is a graphic tool used to display the relationship between two quantitative variables.

6. Tables

- It is used when:
 - Trend is not important
 - Number of values are small
 - To complement other data presentation format

7. Two-way Tables

- Relationships between two categorical variables can be shown through a two-way table.
- It is also known as a contingency table, cross-tabulation table, or a cross-classifying table.

Inferential Statistics

Hypothesis Testing

Hypothesis

- It is an assumption or an idea that proposes a tentative explanation about a phenomenon or a narrow set of phenomena observed in the natural world.
- There are two kinds of hypotheses:



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- A. **Null Hypothesis** (H_o) – is the hypothesis that must always express the idea of non-significance of difference, lack of association or relationship between variables. The rejection of this hypothesis implies the acceptance of the alternative hypothesis.
- B. **Alternative Hypothesis** (H_a) – is the opposite of the null hypothesis. It specifies the existence of a difference (or that one group is better than the other), association or relationship between the dependent and independent variables. The acceptance of the null hypothesis implies the rejection of this hypothesis.

Examples:

- 1. **Title:** The Effects of Online Gaming on the Academic Achievements of Freshman High School Students
 H_o : Online gaming has **no significant effects** on the academic achievements of high school freshman students. (non-directional hypothesis/two-tailed test)
 H_a : Online gaming **has negative effects** on the academic achievements of HS freshman students (directional hypothesis/one-tailed test)
 - 2. **Title:** Correlation between Competencies in English and Mathematics
 H_o : There is **no significant correlation** between competencies in English and Mathematics.
 H_a : There is **a significant correlation** between competencies in English and Mathematics.
- Since H_o testing is based on sample means, not population means, there is a possibility of making an error in rejecting or failing to reject H_o .

Decision Errors		
	H_o is true	H_o is false
Accept	Correct decision	Type II error
Reject	Type I error	Correct decision

- **Type I error** (alpha error or false positive) – rejecting a true null hypothesis
- **Type II error** (beta error or false negative) – accepting a false null hypothesis

Problem Solving

*For more problem-solving exercises, you may refer to the last page of this material.

- **Problem** – a task that requires the learner to reason through a situation that will be challenging but not impossible
- **Exercise** – provides practice in using algorithms and maintaining the basic facts.
- **Problem Solving** – encompasses exploring, reasoning, strategizing, estimating, conjecturing, testing, explaining, and proving.
- **Exercise/Drill** – An exercise/drill is a set of number sentences intended for practice in the development of a skill.

Polya’s Four-Step Process for Problem Solving

- The most famous study of problem-solving techniques was developed by **George Polya**.

- He is known to be “The Father of Problem-Solving”.

Steps in Problem-Solving:

- 1. **Understand the problem (find it out)** – Read the problem, if possible, several times, and analyze it carefully.
 - 2. **Devise a plan (think)** – Devise what appropriate strategy to attack a problem.
 - 3. **Carry out the plan (solve it)** – After knowing a strategy, carry out your plan of attack.
 - 4. **Look back (verification)** – Check if your answer is reasonable.
- The following are suggested problem-solving strategies, which can be used in Step 2 of Polya’s Problem Solving Process.
- Guess and check
 - Look for a pattern
 - Make an orderly list
 - Draw a picture or diagram
 - Solve a simpler problem
 - Use a model
 - Work backwards
 - Use direct reasoning
 - Use a formula
 - Solve an equation
 - Use common sense

Approaches to Problem Solving

A. Deductive Reasoning

- In deductive reasoning, one arrives at a specific conclusion based on general assumptions, procedures, or principles.
- It is the process of reasoning to a specific conclusion from general statement.
- Examples:
 - A quadrilateral has four sides. A square has four sides. Therefore, a square is a quadrilateral.

B. Inductive Reasoning

- In inductive reasoning, one makes generalizations based on individual instances or specific observations.
- Often used by mathematicians and scientists to predict answers to complicated problems.
- Examples:
 - $1 + 3 = 4$
 - $5 + 7 = 12$
 - $11 + 13 = 24$
 - Therefore, the sum of any two odd numbers is an even number.

Routine and Non-routine Problem

Routine Problem

- Using at least one of the four arithmetic operations and/or ratio to solve problems.
- Example: A sales promotion in a store advertises a jacket regularly priced at Php1255.98 but now selling for 20% off the regular price. The store also waives the tax. You have Php1000 in your pocket. Do you have enough money to buy the jacket?

Non-routine Problem

- May be solved in many different ways or strategies and may have more than one answer or solution.
- Mrs. Rivera wishes to tile her classroom floor with square tiles. She wants to use whole tiles, without cutting any pieces. The rectangular floor has dimensions 8.4 meters by 7.2 meters. What is the minimum number of whole identical



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square tiles required and what are the dimensions of each tile?

Math Investigation

What is a mathematical investigation?

- Mathematical investigation refers to the sustained exploration of a mathematical situation. It distinguishes itself from problem solving because it is open-ended.
- Mathematical investigation is the use of mathematical processes to pursue an inquiry about a situation which must be understood or a set of data which must be organized.
- These are activities that involve exploration of open ended mathematical situations. The student is free to choose what aspects of the situation he or she would like to do and how to do it. The students pose their own problem to solve and extend it to a direction they want to pursue. In this activity, students experience how mathematicians work and how to conduct mathematical research.

Standard exercises

- These are activities with clearly defined procedure/strategy and goal. Standard exercises are used for mastery of a newly learned skill – computational, use of an instrument, and even new terms or vocabulary. These are important learning activities but must be used in moderation.

Problem-solving activities

- These are activities involving clearly defined goals but the solutions or strategies are not readily apparent. The student makes a decision on the latter.
- If the students already know how to solve the problem, then it is no longer a problem. It is an exercise.
- It is said that problem solving is at the heart of mathematics.

Mathematical Investigation vs Problem Solving

- Mathematical investigation is a **divergent (open)** activity, i.e. there are no known outcomes either for the students or the teacher.
- Problem solving is a **convergent (closed)** activity where students have to reach a solution to a problem.

Processes Involved in Math Investigation

Mathematical investigations involve the following processes:

1. Exploration
2. Gathering and recording data
3. Looking for patterns
4. Making and testing conjectures
5. Testing the conjecture and then refine it and test again
6. Explaining and proving results
7. Making generalizations
8. Extending the problem
9. Communicating findings

It further involves:

- **Communication Processes** (e.g. explaining, talking, agreeing, questioning)
- **Reasoning Processes** (e.g. analyzing, recognizing patterns, conjecturing, proving)
- **Recording Processes** (e.g. drawing, writing, listing, graphing, symbolizing)

- **Operational Processes** (e.g. data collecting, sorting, ordering, classifying)

Basic Steps in Mathematical Investigation

1. Explore the investigation
2. Gather and record data
3. Look for patterns
4. Make conjectures (guesses)
5. Test the conjecture and then refine it and test it again
6. Explain or prove results
7. Make generalizations
8. Extend the problem
9. Communicate findings

Conjectures

- **Conjectures** are inferences or judgments based on inconclusive or incomplete evidence. (American Heritage Dictionary, 2006)
- These are statements, opinions or conclusions based on guesswork.
- An educated guess
- An unproven statement that is based on observation
- A statement believed to be true, but not proven
- A generalized pattern identified by using inductive reasoning
- Not all conjectures have the same mathematical reach and not all of them are true.
 - Some, refer to relatively simple and self-evident facts.
 - Others, involve a strong element of the unexpected.
 - Still others, are particularly important because from them it is possible to derive many conclusions.

Testing and Proving Conjectures

- Once formulated, a conjecture must go through several tests.
- A single counterexample is enough to reject it. A counterexample is a specific case for which a conjecture is false.
- A conjecture is true if it is true for all cases.
- To show that a conjecture is true for all cases, we must prove it.

Example Investigation:

Consider a non-palindromic three-digit number. Reverse the digits and subtract the smaller from the larger. Record the difference. Reverse the digits of the recorded result and add this to the difference. Investigate.

**Note: A non-palindromic (not palindromic) number is a number when reversed becomes a different number.*

1. Exploration

Read the investigation, discuss what the investigation is all about and what you need to do.

2. Gather and Record Data

$$\begin{aligned} 792 - 297 &= 495 \\ 495 + 594 &= 1089 \end{aligned}$$

$$\begin{aligned} 543 - 345 &= 198 \\ 198 + 891 &= 1089 \end{aligned}$$

$$\begin{aligned} 921 - 129 &= 792 \\ 792 + 297 &= 1089 \end{aligned}$$



Midterm Examinations

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HAU School of Computing
First Year | Bachelor of Science in
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$482 - 284 = 198$
 $198 + 891 = 1089$

$691 - 196 = 495$
 $495 + 594 = 1089$

And so on.

3. Look for patterns

Discuss what you have observed and what patterns you have seen.

4. Make conjectures

Based on the data collected and the patterns observed, make a conjecture. Possible conjectures for the above patterns are as follows:

- a. The final result is always 1089.
- b. After the subtraction step, the middle digit is always 9 and the result is divisible by 9.

5. Gather more data to test conjectures

$683 - 386 = 297$
 $297 + 792 = 1089$

$312 - 213 = 99$
 $198 + 891 = 1089$

$813 - 318 = 495$
 $495 + 594 = 1089$

$756 - 657 = 99$
 $99 + 99 = 198$

$221 - 122 = 99$
 $99 + 99 = 1089$

And so on.

6. Revise conjecture

From the data we have again gathered, we can revise our initial conjectures as follows:

If the first and last digits of the given non-palindromic three-digit number are consecutive integers, the difference is 99 and the sum is 198.

7. Verify conjectures

$887 - 788 = 99$
 $99 + 99 = 198$

$993 - 399 = 594$
 $594 + 495 = 1089$

$711 - 117 = 594$
 $594 + 495 = 1089$

$766 - 667 = 99$
 $99 + 99 = 198$

$661 - 166 = 99$
 $495 + 594 = 1089$

8. Organize Data

Numbers that have resulted in a difference of 99 and a sum of 198 are: 756, 221, 887, 766.

Look carefully at these numbers and see if there is anything about them which is common. Look at the digits. Make a conjecture.

Note that the first and third digits differ by 1.

Organize the data by grouping numbers with the same difference between the first and the third digits:

Digits differ by 2	543, 482	Difference = 198; Sum = 1089
Digits differ by 3	683, 471	Difference = 297; Sum = 1089
Digits differ by 4		
Digits differ by 5	792, 691, 813, 661	Difference = 495; Sum = 1089
Digits differ by 6	993, 711	Difference = 594; Sum = 1089
Digits differ by 7	922	Difference = 693; Sum = 1089
Digits differ by 9		

9. Revise conjectures (generalizations)

- a. The final result is 1089 unless the first and last digits differ by 1 in which case it is 99.
- b. After the subtraction step, the middle digit is always 9 and the other two digits are obtained by multiplying the difference of the first and the third digits, as listed in the table, by 9.
 - For example, consider the number 471: the difference between 4 and 1 is 3. If we multiply 3 by 9, we get 27. Therefore, after the subtraction, the result is 297.

10. Verify

At this point, the important conjecture to verify is the second one and this is done by choosing numbers that have a digit difference of 4, or 9 as there were none of these in the data gathered.

$955 - 559 = 396$
 $396 + 693 = 1098$

955:
 $9 - 5 = 4$
 $4(9) = 36$
After subtraction: 396

$682 - 286 = 396$
 $396 + 693 = 1098$

682:
 $6 - 2 = 4$
 $4(9) = 36$
After subtraction: 396

$970 - 079 = 891$
 $891 - 198 = 693$

970:
 $9 - 0 = 9$
 $9(9) = 81$
After subtraction: 891

Therefore, conjecture is verified.

11. Extend the problem

Consider two-digit numbers.

$95 - 59 = 36$	$36 + 63 = 99$
$86 - 68 = 18$	$18 + 18 = 99$
$72 - 27 = 45$	$45 + 54 = 99$
$63 - 36 = 27$	$27 + 72 = 99$
$21 - 12 = 09$	$09 + 90 = 99$

Conjectures:

- a. The difference between a two-digit number and its reverse is the product of the difference of its two digits and 9.
- b. The sum is always 99.



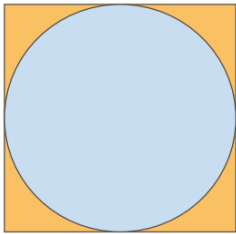
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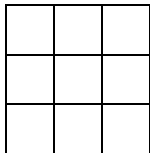
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Problem-Solving Exercises:

- 1. I am thinking of a two-digit number:
 - It is odd.
 - Its tens digit is even.
 - It is prime.
 - The sum of its digits is 11.
 - The product of its digits is 24.What is the number?
- 2. If $F_1 = 1$, what is the 20th Fibonacci number?
- 3. The sum of three consecutive odd integers is 75. Find the largest integer.
- 4. A restaurant has only rectangular tables that can seat 6 people each, with a maximum of 2 people on each side. Find the minimum number of tables that must be joined to form a single long table for a party of 18 people. Show or illustrate how this can be done.
- 5. What is the sum of the first 100 numbers?
- 6. The figure shows a square that is inscribed by the circle with radius 10cm. Find the area of the shaded region.



- 7. If Holee can solve a problem set in 3 hours, and Angelo can answer the same set of questions in 6 hours, how many hours can they finish answering the problem set if they work as a team?
- 8. Teddy invested part of his savings in 6% interest and the rest at 7.5%. The annual interest on two investments totals P6,825. How much money is invested at each rate if the total investment is P100,000?
- 9. Each of four siblings (Edmund, Genalyn, Madelyn, and Sonia) bought four different cars. One chooses Honda, a Mitsubishi, a Toyota, and a Suzuki car. From the following clues, determine which sibling bought which car.
 - a. Edmund, living alone, stays next door to his sister who bought the Honda car and very far from his sister who bought the Suzuki car.
 - b. Genalyn, also living alone, is younger than the one who bought the Mitsubishi car and older than her sibling who bought the Toyota car.
 - c. Madelyn did not like Toyota and Suzuki cars. But she and her sibling, who bought the Toyota car, live in the same house.
- 10. Put the numbers 1 to 9 into the square so that all rows, columns and diagonals add to the magic number.



- 11. Mercy has a certain amount in her bank account on Friday morning. During the day she writes a check for \$ 24.50, makes an ATM withdrawal of \$80 and deposits a check for \$ 235. At the end of the day, she sees that her balance is \$ 451.25. How much money did she have in the bank at the beginning of the day?
- 12. Trina’s father is 36. He is 16 years older than four times Trina’s age. How old is Trina?

Mathematical Investigation Exercises:

- 1. How many squares of all sizes are in a 12 x 12 checkboard? Investigate.
- 2. Investigate the sum of two to five consecutive counting numbers.
- 3. How many rectangles of all sizes are in a 6 x 6 board? Investigate.



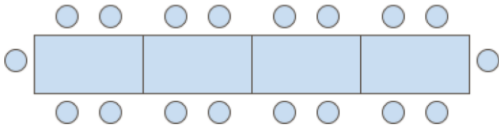
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ANSWER KEY

1. 83
2. Using the Fibonacci formula:
$$F_n = \frac{1}{\sqrt{5}} \left[\left(\frac{1 + \sqrt{5}}{2} \right)^n - \left(\frac{1 - \sqrt{5}}{2} \right)^n \right]$$
$$F_{20} = \frac{1}{\sqrt{5}} \left[\left(\frac{1 + \sqrt{5}}{2} \right)^{20} - \left(\frac{1 - \sqrt{5}}{2} \right)^{20} \right]$$
$$F_n = 6765$$
3. Let $x = 1^{\text{st}}$ odd integer (smallest);
 $x + 2 = 2^{\text{nd}}$ odd integer;
 $x + 4 = 3^{\text{rd}}$ odd integer (largest).
$$x + (x + 2) + (x + 4) = 75$$
$$3x + 6 = 75$$
$$3x = 75 - 6$$
$$\frac{3x}{3} = \frac{69}{3}$$
$$x = 23$$

Therefore, the largest odd integer is $23 + 4 = 27$.
4. The tables must be arranged as illustrated below:


Therefore, four (4) tables must be joined to form a single long table for a party of 18 people seated as shown in the above figure.
5. Look for a pattern – note that there is actually a pattern such that:
$$1 + 100 = 101$$
$$2 + 99 = 101$$
$$3 + 98 = 101$$

And so on. There are 50 pairs whose sum is 101. Therefore, the sum of the first 100 numbers is:
$$50 (101) = 5,050$$
6. In a square that is inscribed by a circle, the diameter of the circle is equal to the side of the square. Therefore, from the given figure:
$$d = 2r$$
$$d = \text{diameter};$$
$$r = \text{radius};$$
$$d = 20 = \text{side}$$
$$s = \text{side}$$

Note that: Area of the shaded region = area of the square – area of the rectangle
$$= s^2 - \pi r^2$$
$$= 20^2 - 3.14(10)^2$$
$$= 86cm^2$$
7. Let $t =$ no. of hours it takes Holo & Angelo to finish answering the problem set together. Then:
$$\frac{1}{1} + \frac{6}{1} = \frac{7}{t}$$
$$6t \left[\frac{1}{1} + \frac{6}{1} \right] = 6t \left[\frac{7}{t} \right]$$

8. Formula for simple interest:
$$I = Prt$$

Where: $I =$ simple interest
 $P =$ principal
 $r =$ rate of interest, %
 $t =$ time in years

Let:
 $P_1 =$ amount invested at 6%
 $P_2 = 100,000 - P_1 =$ amount invested at 7.5%
Equation:
$$I^1 + I^2 = 6825$$
$$P_1(0.06)(1) + (100,000 - P_1)(1) = 6,825$$
$$0.06P_1 + 7500 - 0.075P_1 = 6,825$$
$$-0.015P_1 = 6,825 - 7,500$$
$$\frac{-0.015P_1}{-0.015} = \frac{-675}{-0.015}$$
$$P_1 = 45,000 \text{ is the amount invested at 6\%}$$
$$P_2 = 55,000 \text{ is the amount invested at 7.5\%}$$
9. Edmund bought a Mitsubishi car.
Genalyn bought a Suzuki car.
Madelyn bought a Honda car.
Sonia bought a Toyota car.
10.

8	3	4
1	5	9
6	7	2
11. We need to find the money in Mercy's bank account at the beginning of the day on Friday.
She took \$24.50 and put \$235. She ended up \$451.25.
$$\$451.25 - \$235 - \$80 + \$24.50 = \$320.75$$
12. To get from Trina's age to her father's age, we multiply Trina's age by four and add 16. Working backwards means we start with the father's age, so subtract 16 and divide by 4.
$$36 - 16 = 20$$
$$20 \div 4 = 5$$

To check, Trina is 5 years old. Her father's age is $4(5) + 16 = 36$. This is correct.